1. Identity and decision problem

Musanze Harvest Intelligence Service

Identity	Confirmed details
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Institution / course	INES Ruhengeri / SWE 3513 - Artificial Intelligence
Assessment / signature	Individual Assignment 2 / AI-A2-7343
Deadline	19 October 2026, 5:00 PM CAT

A Musanze potato cooperative needs to consolidate inconsistent farm records into estimates that support harvest collection and field visits. The field officer remains the decision maker.

Task	Question and output
Regression	How many tons will this farm harvest? Output: predicted_yield_tons.
Classification	Is synthetic yield per hectare below the base-data median? Output: high-risk label and probability.
Clustering	Which farms have similar measured conditions? Output: cluster membership.

Ethical boundary

This is a classroom prototype trained entirely on synthetic data. It must not replace an agronomist or decide access to finance, insurance or public services. A high-risk label is a dataset definition, not a diagnosis of crop disease or a verified forecast of farm failure.

AI-assisted implementation and draft explanations are disclosed in evidence/AI_USE.md. The student must personally review, rewrite and defend the interpretations in docs/PERSONAL_REVIEW.md. The declaration is unsigned and the authentic demonstration is pending.

2. Data and reproducibility

Column	Meaning / range
rainfall_mm	600 to 1400; numeric
avg_temp_c	12 to 26; numeric
soil_moisture_pct	20 to 80; numeric
fertilizer_kg	0 to 250; numeric
field_size_ha	0.2 to 4; numeric
cooperative_support	0 to 1; binary
season	A or B
yield_tons	Synthetic harvest target; minimum 0.50 tons
high_risk	1 if pre-missingness yield/ha is below base median; otherwise 0

Seed = last four registration digits = 7343. numpy.random.default_rng(7343); base rows = 240 + (7343 % 41) = 244. Labels are created before missingness, using unrounded generated values. The median is part of the specified synthetic label generator, not a fitted prediction-time feature.

Yield = 0.004 rainfall + 0.06 moisture + 0.02 fertilizer + 1.30 field size - 0.25 abs(temperature - 18) + 0.70 support + season effect + N(0, 0.80). Season A adds 0.50; B subtracts 0.20. Clip at 0.50.

Raw rows 247; duplicates 3; cleaned rows 244. Exactly 12 unique positions per missingness column are sampled separately (overlap is allowed). The first three corrupted records are appended as exact duplicates. Clean missing counts: moisture 12, fertilizer 12; raw counts: moisture 12, fertilizer 12.

CSV floats have four decimals; UTF-8 with explicit LF line endings. SHA-256 hashes final CSV bytes:

19a1558643adbce326107d7fcfb9df4a
8cd48cf745f2af43b8e07af6b6975d62

Two complete pipeline executions produced identical CSV fingerprint, metric JSON files, cluster report and test-prediction CSV bytes. See evidence/reproducibility.json. Missing predictors are preserved until training-only median imputation.

3. Pipeline design and leakage controls

After removing duplicate records, a fixed stratified 75/25 split produces 183 training and 61 testing farms; random_state=7343. Original zero-based CSV positions remain row IDs. Duplicate removal precedes the split so copies cannot appear on opposite sides.

Stage	Transformation / control
Raw CSV -> validation	Exact schema, numeric bounds, categorical values, missingness and duplicate evidence.
Clean -> split	No imputation, scaling or encoding is fitted before splitting.
Numeric branch	Training medians -> StandardScaler fitted on training inputs. Binary support is scaled with other numeric inputs.
Season branch	OneHotEncoder fitted on training season values; inference accepts A/B only.
Model -> saved pipeline	Each candidate has an independent ColumnTransformer. Final joblib pipelines retain their fitted transformers.

Only rainfall, temperature, moisture, fertilizer, field size, cooperative support and season are inputs. An explicit whitelist excludes yield_tons, high_risk and any yield-per-hectare column. Clustering uses the same measured-input whitelist and training farms only.

Regression candidates: mean DummyRegressor, LinearRegression, and RandomForestRegressor (200 trees). Classification: most-frequent DummyClassifier, LogisticRegression and RandomForestClassifier (200 trees). Forests use one worker; random states are fixed. KMeans tests K=2..6 with n_init=20.

Selection uses lowest test RMSE for regression and highest test F1 for classification (recall breaks ties). The PDF permits selection on test evidence, but this makes the selected score optimistic as an estimate of future performance. Future work should use train/validation selection and a separate untouched test set.

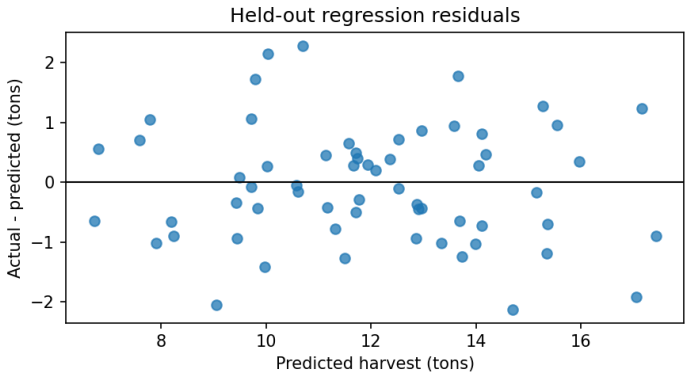
Training -> artifacts/models -> shared Predictor -> CLI and FastAPI. Inference never retrains; the API loads saved trusted pipelines at startup. Tests verify stored imputer medians against training medians and confirm feature exclusion.

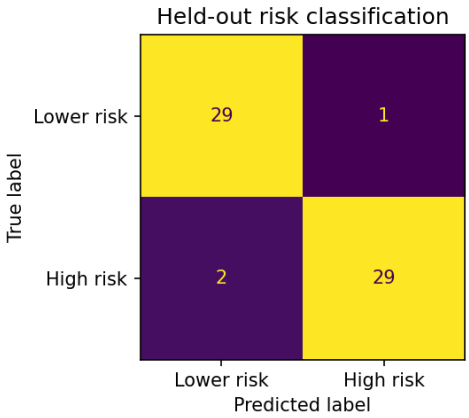
4. Regression and classification

Regressor	MAE	RMSE	R squared
DummyRegressor	2.1753	2.7148	-0.0147
LinearRegression	0.7967	0.9709	0.8702
RandomForestRegressor	0.9501	1.2217	0.7945

Classifier	Precision	Recall	F1	ROC-AUC
DummyClassifier	0.0000	0.0000	0.0000	0.5000
LogisticRegression	0.9667	0.9355	0.9508	0.9892
RandomForestClassifier	0.8966	0.8387	0.8667	0.9769

Selected: LinearRegression (lowest RMSE); LogisticRegression (highest F1). Positive class = 1, probability threshold = 0.5. All candidates share the same held-out farms.





Residual sign in the plot is actual minus predicted; the CSV uses predicted minus actual. Confusion-matrix rows are actual classes, columns predicted classes. ROC-AUC uses probabilities and is reported only with both true classes.

CSV row 2: actual 12.9863 tons; predicted 10.7003; predicted minus actual -2.2860 tons. This is the largest absolute test error. Such an error could distort transport and storage planning. Missing measurements, noise and the linear model's inability to represent the temperature kink are plausible contributors, not a proven explanation for this row.

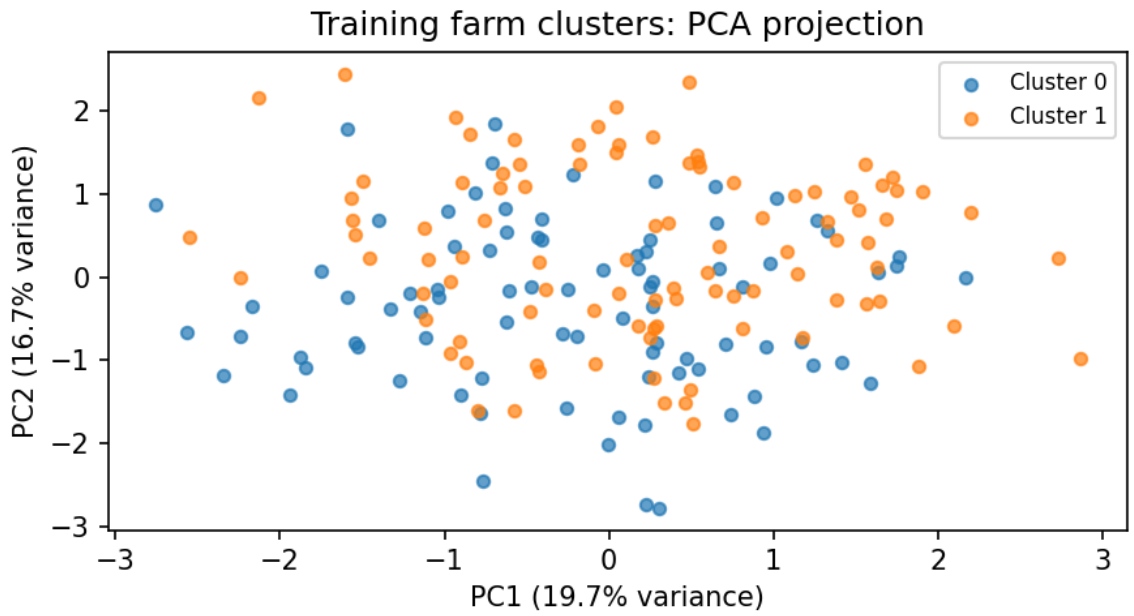
CSV row 28: actual risk 1, predicted 0, probability 0.4500. This false negative could cause a field officer to overlook a farm needing assessment.

These are AI-assisted factual interpretation drafts. Student review and personal wording are required before final submission; see docs/PERSONAL_REVIEW.md.

5. Clustering and operational meaning

K	Training silhouette
2	0.15659
3	0.13650
4	0.13602
5	0.13809
6	0.13716

Selected K=2 maximizes silhouette among the five candidates. The low best score (0.157) indicates weak separation. Here the groups split exactly by cooperative support; other mean conditions are similar. Silhouette does not establish agronomic validity. Cluster IDs are arbitrary; no target enters fitting.



The PCA projection is fitted on training inputs; axes show the explained variance of each principal component. Separation in two dimensions may conceal overlap in the full feature space.

Training group 0: 86 farms; mean rainfall 1017.9 mm, temperature 19.7 C, field size 2.01 ha and support fraction 1.00. Use these measured similarities for field-visit planning, subject to officer review.

Training group 1: 97 farms; mean rainfall 990.0 mm, temperature 19.5 C, field size 2.01 ha and support fraction 0.00. Use these measured similarities for field-visit planning, subject to officer review.

Profiles use observed numeric means (missing measurements omitted) and include season proportions in the JSON. These groups may help coordinate visits or compare similar farms. They do not identify causal interventions. Student: choose a measured cluster and rewrite its operational interpretation in PERSONAL_REVIEW.md.

6. API contract and actual testing

Endpoint	Contract
GET /health	200: status and models_ready. Missing artifacts produce degraded health.
GET /model-info	200: selected models, features, K, seed, verification code and fingerprint; 503 if unavailable.
POST /predict	Seven required bounded fields -> yield, risk label, probability, cluster and verification code. Invalid input -> 422 with field details; unavailable models -> 503.
GET /docs	Swagger UI; /openapi.json describes typed request and response.

Valid request: {"rainfall_mm":1000,"avg_temp_c":18,"soil_moisture_pct":50,"fertilizer_kg":120,"field_size_ha":2,"season":"A","
cooperative_support":1}

Actual CLI response: {
"predicted_yield_tons": 12.4816924638537,
"predicted_high_risk": 0,
"high_risk_probability": 0.33931241711599996,
"cluster": 0,
"verification_code": "AI-A2-7343"
}

Invalid sample combines rainfall 1500, temperature "warm" and season C. CLI returned exit 2 with validation details. API tests exercise missing fields, invalid values and malformed JSON (HTTP 422). Numeric strings, boolean support, NaN and extra target fields are rejected by shared strict Pydantic validation.

Executed evidence

28 passed, 1 warning in 17.36s

pytest checks valid inference, missing/type/range/season/extra/nonfinite failures, reproducible bytes, 247/244 row counts, 12 missing values per cleaned column, signature, saved model loading, wrong seed rejection, leakage controls, train-only imputation, CLI behavior and FastAPI responses. Full raw stdout/stderr is in evidence/execution_log.json.

FastAPI TestClient verified health, model-info, docs, valid prediction, missing-field validation and malformed JSON. Explicit endpoint responses are saved in evidence/api_checks.json. No authentic video or student signature is claimed. No live browser interaction, load test or real agricultural validation is claimed here.

7. Responsible AI model card

Aspect	Scope and limitation
Intended user / use	Cooperative field officers; classroom harvest planning and prioritization support.
Data provenance	244 synthetic base farms. No actual farm measurements, farmer identities or consented field outcomes.
Target meaning	High risk means below-median synthetic tons/ha, not disease, poverty or an independently validated failure event.
Performance scope	One fixed split. Test evidence is also used for model selection. Small sample and random missingness constrain reliability.
Risks	Overconfident estimates, unfair service allocation, synthetic correlations, measurement errors and failure under distribution shift.
Prohibited use	Automatic finance, insurance or public-service access decisions; replacing an agronomist.
Human oversight	Confirm farm measurements, inspect uncertain cases and obtain agronomist advice before action. Avoid punitive interpretations of risk labels.
Security / privacy	No authentication or production rate limiting. Bind locally. Load only trusted joblib files, which can execute code. Collect minimal farmer data in any future deployment.

Improvement plan: gather representative real seasonal data with consent, verify measurement quality, use separate validation/test sets, test subgroup performance, quantify harvest uncertainty and calibrate risk probabilities. Compare richer temperature features under training-only cross-validation. Monitor drift and document retraining decisions.

Personal reflection awaiting student review

The factual error and cluster analyses are AI-assisted drafts, not invented student reflections. Alain Sauverain ISHEMA must review the editable worksheet, explain the actual results in personal wording, and confirm that the declaration matches the work performed. The required continuous personal video and handwritten card remain outstanding.

Source: supplied eight-page INES Ruhengeri SWE 3513 Individual Assignment 2 PDF, issued 16 October 2026. All numerical claims in this report are read from the accompanying generated artifacts.